

Applied A.I. Solutions

Data Visualization Techniques

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DATA VISUALIZATION TECHNIQUES

Parameters and Calculated Fields

Parameters

- fields of a data source
- mainly used in calculations as variable
- useful for what-if analysis
- way of letting the user specify a value

A red arrow originates from the right side of the list, specifically pointing towards the 'way of letting the user specify a value' item, and points down to the variable 'X' in the equation below.
$$f(x) = a \text{ X } + b$$

Parameters – cont'd

Parameters are useful when you want:

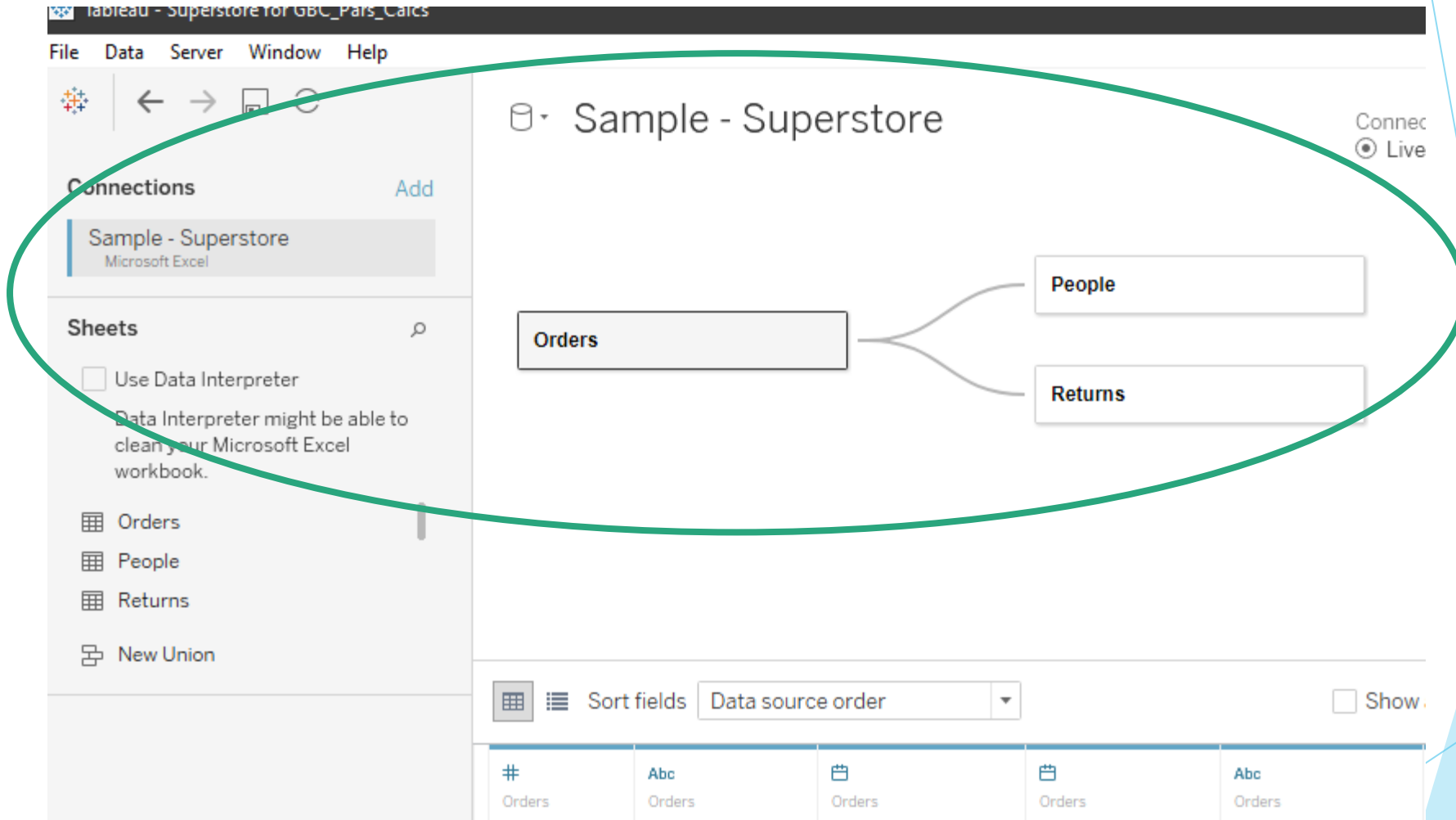
- to add interactivity and flexibility to a report, or
- to experiment with what-if scenarios

In **Tableau**, you need to **tie the parameters to the view**:

- Use parameters in calculations and **calculated fields** used in the view
- Display the parameter control in the view for **users** to select parameters

<https://help.tableau.com/current/pro/desktop/en-us/changing-views-using-parameters.htm>

1



The screenshot shows the Tableau Desktop interface with the title bar 'Tableau - Superstore for GBC_Pars_Cards'. The menu bar includes 'File', 'Data', 'Server', 'Window', and 'Help'. On the left sidebar, the 'Connections' pane shows 'Sample - Superstore' (Microsoft Excel) selected. Below it, the 'Sheets' pane lists 'Orders', 'People', 'Returns', and 'New Union'. A green oval highlights the 'Connections' and 'Sheets' panes. The main workspace, titled 'Sample - Superstore', displays a data model diagram where 'Orders' is connected to 'People' and 'Returns'. The bottom of the interface shows a 'Sort fields' dropdown set to 'Data source order' and a 'Show' checkbox. Below this is a table with five columns, each labeled 'Orders'.

#	Abc	Abc	Abc	Abc
Orders	Orders	Orders	Orders	Orders

1

File Data Server Window Help

Sample Superstore

Connections Add

Sample ...erstore
Microsoft Excel

Sheets

☐ Use Data Interpreter
Data Interpreter might be able to clean your Microsoft Excel workbook.

Orders
People
Returns
New Union

Sales Commission
Sales Target (US)
☒ Sample - Superstore
New Data Source

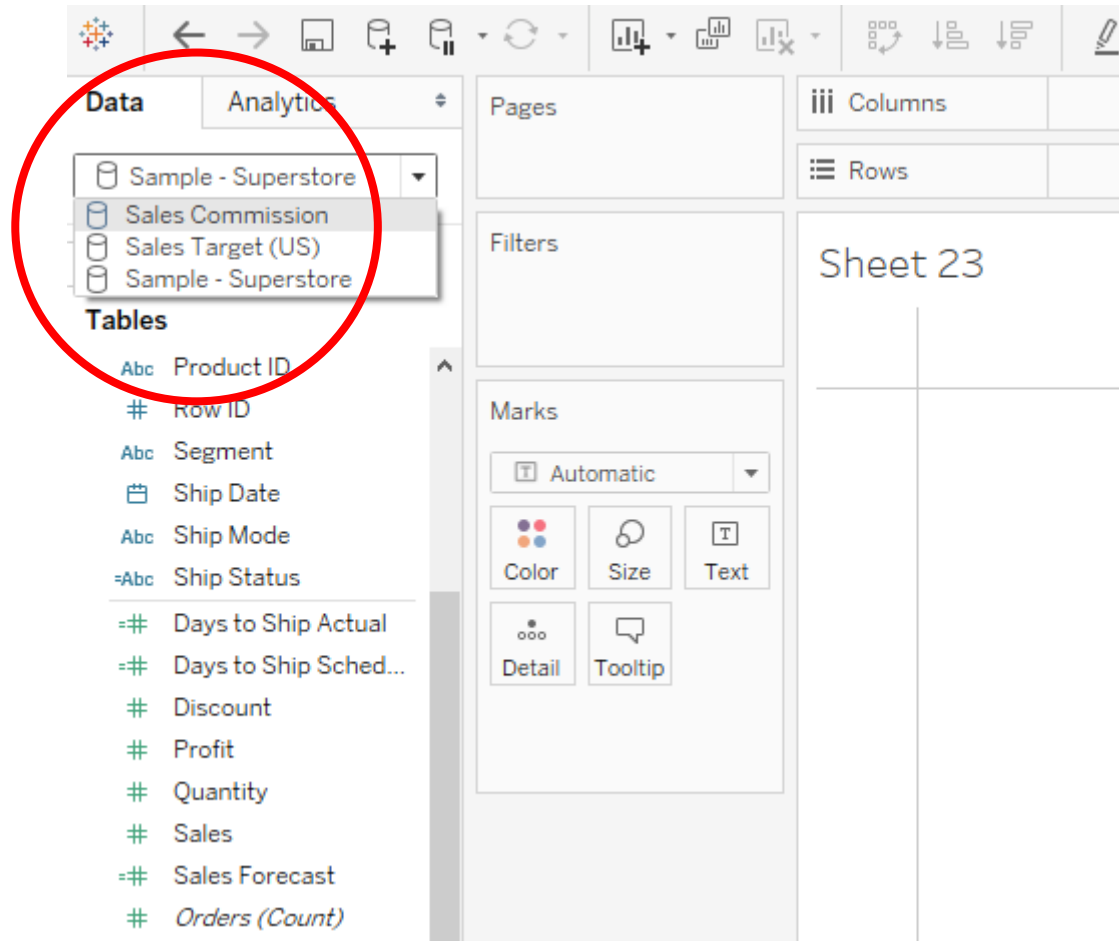
Orders

People
Returns

Sort fields Data source order

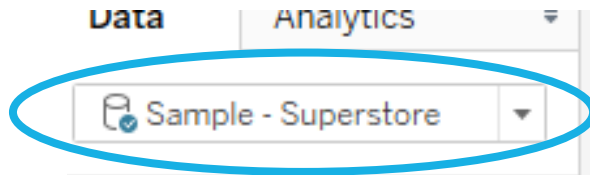
# Orders Row ID	Abc Orders Order ID	Orders Order Date	Orders Ship Date
1	CA-2019-152156	11/8/2019	11/11/20
2	CA-2019-152156	11/8/2019	11/11/20

1

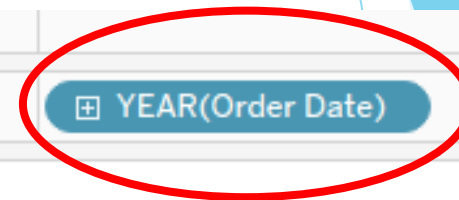


The screenshot shows the Tableau Desktop interface. The 'Data' pane on the left lists several tables: 'Sample - Superstore', 'Sales Commission', 'Sales Target (US)', and 'Sample - Superstore'. A red circle highlights the 'Sales Commission' table. Below this, a list of fields is shown, including 'Product ID', 'Row ID', 'Segment', 'Ship Date', 'Ship Mode', 'Ship Status', 'Days to Ship Actual', 'Days to Ship Sched...', 'Discount', 'Profit', 'Quantity', 'Sales', 'Sales Forecast', and 'Orders (Count)'. The 'Columns' shelf is empty, and the 'Rows' shelf is labeled 'Sheet 23'. The 'Marks' shelf is set to 'Automatic' and includes 'Color', 'Size', 'Text', 'Detail', and 'Tooltip' cards.

2



1

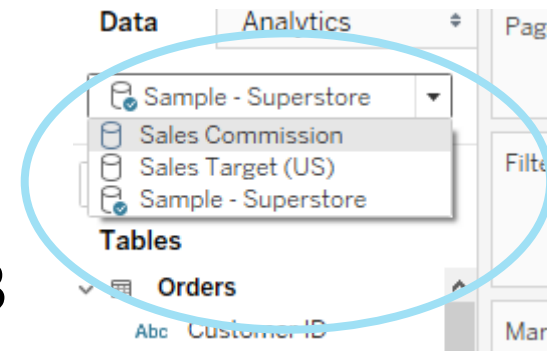


Sheet 23

Year of Ord..

2017	Abc
2018	Abc
2019	Abc
2020	Abc

3



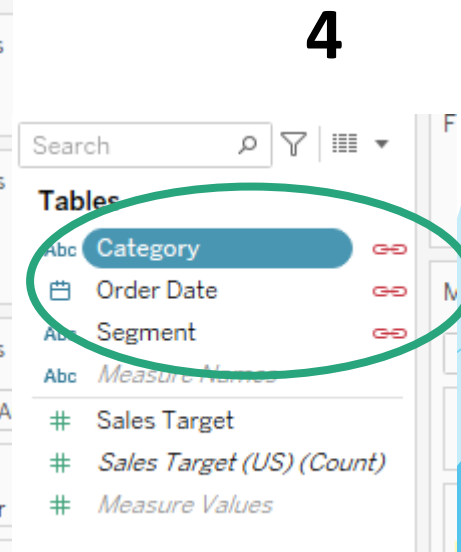
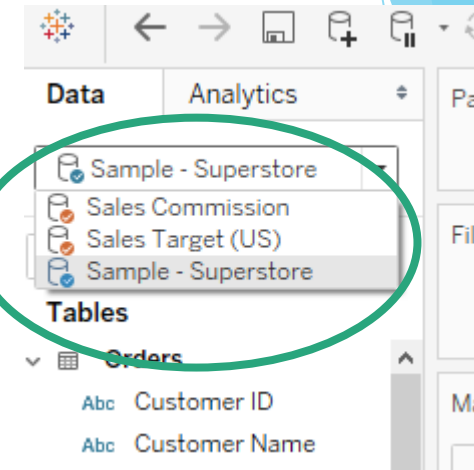
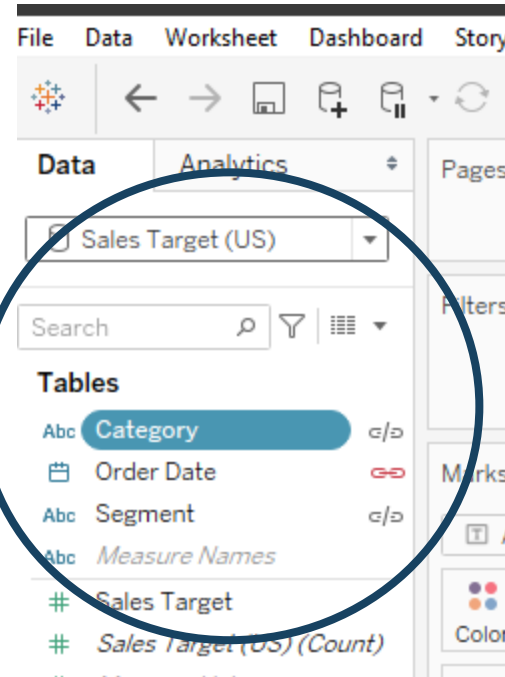
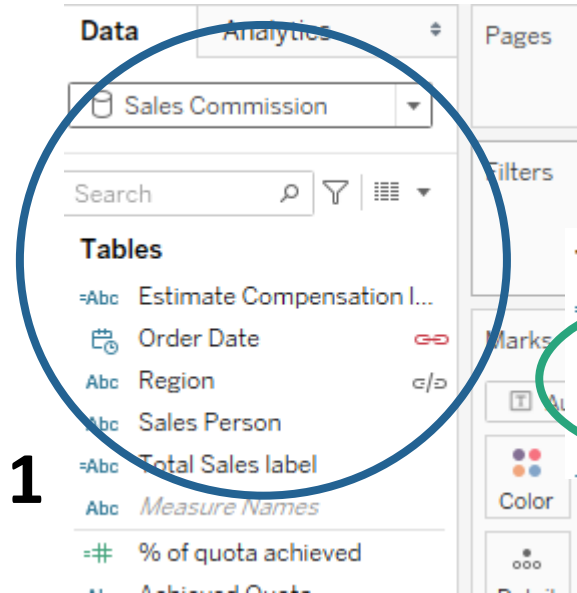
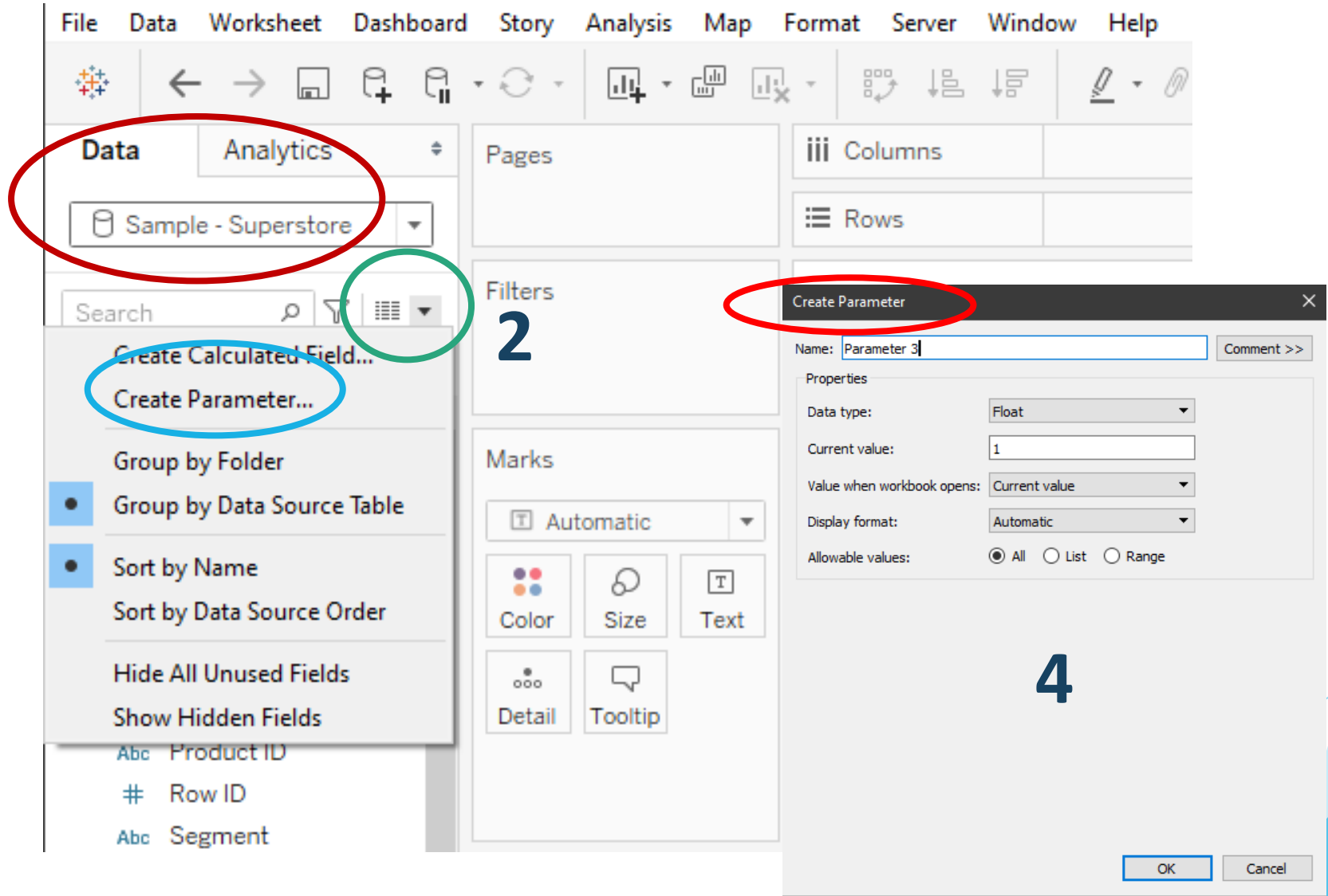


TABLEAU – Parameters

1

3



The screenshot illustrates the process of creating a parameter in Tableau. The interface includes a menu bar (File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, Help) and a toolbar. The 'Data' tab is selected, and the data source is 'Sample - Superstore'. The 'Data' menu is open, showing options like 'Create Calculated Field...', 'Create Parameter...', 'Group by Folder', 'Group by Data Source Table', 'Sort by Name', 'Sort by Data Source Order', 'Hide All Unused Fields', and 'Show Hidden Fields'. The 'Create Parameter...' option is highlighted. The 'Create Parameter' dialog box is open, showing the following settings:

- Name: Parameter 3
- Comment: >>
- Properties:
 - Data type: Float
 - Current value: 1
 - Value when workbook opens: Current value
 - Display format: Automatic
 - Allowable values: ☒ All ☐ List ☐ Range

The 'OK' and 'Cancel' buttons are at the bottom right of the dialog box.

1

Edit Parameter [Select Column 1 Heading] ✕

Name: Comment >>

Properties

Data type: 2

Current value:

Value when workbook opens:

Display format:

Allowable values: ☐ All ☒ List ☐ Range

List of values

Value	Display As
None	None
Customer Name	Customer Name
Customer Segment	Customer Seg...
Region	Region
Category	Category
Segment	Segment
Add	

☒ Fixed Add values from ▶

☐ When workbook opens None ▼

Clear All

OK Cancel

3

4



Calculation1 Sample - Superstore

Column1 Category Sample - Superstore

```
CASE [Select Column 1 Heading]
WHEN 'Customer Name' THEN [Customer Name]
WHEN 'Region' THEN [Region]
WHEN 'Category' THEN [Category]
WHEN 'Segment' THEN [Segment]
ELSE ''
END
```

The calculation is valid. 1 Dependency

Apply OK

ABS (number)

Returns the absolute value of the given number.

e: ABS(-7) = 7

ABS

ACOS

AND

ASCII

ASIN

ATAN

ATAN2

ATTR

AVG

BUFFER

CASE

1

2

3

Tables

- # Discount
- # Profit
- # Quantity
- # Sales
- =# Sales Forecast
- # Orders (Count)

▼ **People**

- Abc Person
- # People (Count)

▼ **Returns**

- Abc Returned
- # Returns (Count)

Abc Measure Names

- =# Column 1 Category
- =# Column 2 Category
- =# Profit per Order
- =# Profit Ratio
- =# Row 1 Category
- =# Row 2 Category
- =Abc Sales above Target?
- =# Sales per Customer
- 🌐 Latitude (generated)

Parameters

- # Base Salary
- # Churn Rate
- # Commission Rate
- # New Business Growth
- # New Quota
- Abc Select Column 1 Heading
- Abc Select Column 2 Heading
- Abc Select Row 1 Heading
- Abc Select Row 2 Heading

1

2

3

Context menu options:

- Add to Sheet
- ✓ Show Parameter
- Cut
- Copy
- Edit...
- Duplicate
- Rename
- Hide
- Delete
- Create
- Default Properties
- Folders
- Replace References...
- Describe...

2

Select Column 1 Heading

None

Select Column 2 Heading

None

Select Row 1 Heading

None

Select Row 2 Heading

None



1

Data Analytics

Sample - Supe...

Search

Filters

Tables

Orders

Column 1 Category

Column 2 Category

Customer ID

Customer Name

Location

Country/Region

State

City

Postal Code

Order Date

Order ID

Order Profit

Product

Category

Sub-Category

Product Name

Product ID

Row 1 Category

Row 2 Category

Row ID

Segment

Ship Date

Ship Mode

Parameters

Base Salary

Churn Rate

Commission Rate

New Business Growth

New Quota

Columns

Column 1 Category

Column 2 Category

Sub-Category

Rows

Row 1 Category

Row 2 Category

Marks

Automatic

Color

Size

Text

SUM(Sales)

Region

2

Sheet 22

		Column 1 Category / Column 2 Category / Sub-Category																
Row 1 Cate...	Row 2 Cate...	Furniture				Office Supplies									Technology			
		Bookc...	Chairs	Furnis...	Tables	Applia...	Art	Binders	Envelo...	Fasten...	Labels	Paper	Storage	Suppli...	Access...	Copiers	Machi...	Phones
Consumer	Central	\$12,961	\$42,933	\$9,500	\$20,835	\$9,819	\$3,272	\$39,694	\$2,365	\$422	\$743	\$7,564	\$24,672	\$4,560	\$19,062	\$4,520	\$12,000	\$37,109
	East	\$27,307	\$59,645	\$13,177	\$14,082	\$19,390	\$3,847	\$31,623	\$2,096	\$424	\$1,653	\$8,661	\$24,994	\$8,568	\$22,861	\$30,400	\$30,668	\$51,512
	South	\$8,717	\$26,804	\$10,871	\$24,409	\$8,287	\$2,655	\$18,618	\$1,192	\$276	\$1,868	\$7,603	\$16,882	\$2,124	\$11,652	\$4,440	\$18,339	\$30,846
	West	\$19,648	\$43,480	\$16,072	\$40,607	\$15,324	\$4,478	\$28,227	\$2,119	\$558	\$2,445	\$12,496	\$33,944	\$10,490	\$33,530	\$30,460	\$18,536	\$50,466
Corporate	Central	\$8,390	\$27,990	\$3,144	\$12,563	\$5,225	\$1,479	\$8,880	\$1,236	\$232	\$1,095	\$5,952	\$12,657	\$4,383	\$6,195	\$29,680	\$11,717	\$17,181
	East	\$13,169	\$23,505	\$9,983	\$17,552	\$10,981	\$2,278	\$12,048	\$1,579	\$286	\$509	\$5,975	\$30,968	\$1,849	\$11,748	\$4,220	\$23,892	\$29,866
	South	\$1,154	\$11,367	\$4,505	\$12,620	\$10,124	\$1,332	\$11,828	\$1,471	\$81	\$365	\$3,345	\$11,362	\$6,022	\$10,175	\$4,860	\$12,001	\$19,274
	West	\$11,294	\$36,279	\$7,369	\$28,138	\$10,259	\$3,501	\$18,805	\$1,657	\$184	\$2,133	\$8,611	\$24,804	\$7,181	\$20,072	\$8,070	\$12,667	\$24,833
Home Office	Central	\$2,807	\$14,308	\$2,610	\$5,757	\$8,539	\$1,014	\$8,350	\$1,036	\$123	\$613	\$3,976	\$8,602	\$525	\$8,699	\$3,060	\$3,081	\$18,114
	East	\$3,343	\$13,110	\$5,911	\$7,506	\$3,817	\$1,360	\$9,827	\$701	\$110	\$442	\$5,536	\$15,650	\$343	\$10,424	\$18,600	\$11,546	\$19,237

Select Column 1 Heading

Customer Segment

Select Column 2 Heading

Category

Select Row 1 Heading

Segment

Select Row 2 Heading

Region

Region

Central

East

South

West

3

Sheet 22

Filters

Customer Name

Marks

1

Automatic

Column 1 Category / Column 2 Category / Sub-Category

Furniture

Chairs Furnis.. Tables

Office Supplies

Art Binders Envelo.. Fasten.. Labels Paper Storage Suppli..

Technology

Access.. Copiers Machi..

\$42,933 \$9,500 \$20,835

\$9,819 \$3,272 \$39,694

\$2,365 \$422 \$743

\$7,564 \$24,672 \$4,560

\$19,062 \$4,520 \$12,000

\$59,645 \$13,177 \$14,082

\$19,390 \$3,847 \$31,623

\$2,096 \$424 \$1,653

\$8,661 \$24,994 \$8,568

\$22,861 \$30,400 \$30,668

\$26,804 \$10,871 \$24,409

\$8,287 \$2,655 \$18,618

\$1,192 \$276 \$1,868

\$7,603 \$16,882 \$2,124

\$11,652 \$4,440 \$18,339

\$43,480 \$16,072 \$40,607

\$15,324 \$4,478 \$28,227

\$2,119 \$558 \$2,445

\$12,496 \$33,944 \$10,490

\$33,530 \$30,460 \$18,536

\$50,466

\$27,990 \$3,144 \$12,563

\$5,225 \$1,479 \$8,880

\$1,236 \$232 \$1,095

\$5,952 \$12,657 \$4,383

\$6,195 \$29,680 \$11,717

\$17,181

\$23,505 \$9,983 \$17,552

\$10,981 \$2,278 \$12,048

\$1,579 \$286 \$509

\$5,975 \$30,968 \$1,849

\$11,748 \$4,220 \$23,892

\$29,866

\$11,367 \$4,505 \$12,620

\$10,124 \$1,332 \$11,828

\$1,471 \$81 \$365

\$3,345 \$11,362 \$6,022

\$10,175 \$4,860 \$12,001

\$19,274

\$11,294 \$36,279 \$7,369

\$28,138 \$10,259 \$3,501

\$18,805 \$1,657 \$184

\$2,133 \$8,611

\$24,804 \$7,181 \$20,072

\$8,070 \$12,667

\$24,833

\$2,807 \$14,308 \$2,610

\$5,757 \$8,539 \$1,014

\$8,350 \$1,036 \$123

\$613 \$3,976

\$8,602 \$525 \$8,699

\$3,060 \$3,081

\$18,114

\$3,343 \$13,110

\$5,911 \$7,506

\$3,817 \$1,360

\$9,827 \$701 \$110

\$442 \$5,536

\$15,650 \$343

\$10,424 \$18,600

\$11,546 \$19,237

2

Customer Name

☒ (All)

☒ Aaron Bergman

☒ Aaron Hawkins

☒ Aaron Smayling

☒ Adam Bellavance

☒ Adam Hart

☒ Adam Shillingsburg

☒ Adrian Barton

☒ Adrian Hane

☒ Adrian Shami

☒ Aimee Pixby

☒ South

☒ West

